

P8139 Final Project Proposal: methylation-driven transcriptional silencing in TCGA-KIRC

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1 Cancer type and research question

Cancer type. Clear-cell renal cell carcinoma (ccRCC), TCGA project KIRC. ccRCC is chosen because it is the largest paired-design cohort in TCGA, its molecular biology is dominated by VHL inactivation (a well-characterized positive control), and its single dominant histology limits biological heterogeneity (TCGA Research Network 2013; Kaelin 2007).

Motivating problem. Paired tumor-adjacent-normal methylation analyses on the Illumina 450K platform routinely declare more than half of tested CpGs differentially methylated at $\text{FDR} < 0.05$. The list is statistically valid but biologically uninformative because most differentially methylated CpGs are passenger or cell-composition changes, not regulators of transcription.

Research question. *Which CpG methylation changes drive transcriptional silencing of their cognate gene?* This is the epigenetic counterpart of Knudson’s tumor-suppressor second hit and the operationally relevant question for cancer epigenetics. We address it by integrating paired DNA methylation with paired RNA-seq from the same patients, requiring concordant evidence from two statistical tests before declaring a CpG transcriptionally silencing.

2 Data sources, samples, and study design

Data acquisition. DNA methylation 450K β -values, RNA-seq STAR-Counts (used for $\log_2(\text{TPM} + 1)$), and harmonized clinical metadata are retrieved from the GDC with `TCGAbiolinks::GDCquery` and `GDCdownload` (Bioconductor 3.22; Colaprico et al. 2016).

Samples and design. This is a **paired case-control design** at the methylation level (each patient contributes a primary-tumor -01A aliquot and a solid adjacent-normal -11A aliquot) combined with an **independent case-mostly design** at the integration level (most patients have only the tumor RNA-seq). One -05A “additional new primary” is excluded by pre-registration. Sample counts:

Table 1: TCGA-KIRC cohort sizes after pre-registered exclusions.

Modality	N
DNA methylation 450K (aliquots)	484 (324 tumor + 160 normal)
DNA methylation 450K (patients)	319
Paired methylation set (both -01A and -11A)	160 patients
RNA-seq STAR-Counts (patient $\times$ sample-type keys)	605
Integration set: matched (patient, sample-type) keys	342 (318 tumor + 24 adjacent-normal)

The DM test runs on the 160 paired patients; the methylation–expression integration test runs on the 342-key matched set.

3 DNA methylation preprocessing steps

The 485,577-probe β -value matrix is filtered sequentially, with attrition recorded. Preprocessing is standard and conservative; no probes are added back at any later stage.

Table 2: Preprocessing attrition for DNA methylation.

Step	Probes retained
Baseline (450K array)	485,577
Intersect with Illumina 450K annotation (drop unmapped)	472,651
Drop chr X / chr Y (avoid sex-driven PC2 in EDA)	461,295
Cross-reactive + SNP-overlapping probes (DMRcate; Chen 2013, Peters 2015)	433,595
Drop probes with $> 20\%$ sample missingness	395,325
Complete-case across all 484 samples (paired-test ready)	320,953

β -values are clipped to $[0.001, 0.999]$ and transformed to $M = \log_2(\beta/(1-\beta))$ for variance stabilization in the downstream paired test (Du et al. 2010). RNA-seq counts are converted to TPM and log-transformed as $\log_2(\text{TPM} + 1)$; replicate aliquots within (patient, sample-type) are mean-aggregated.

4 Analysis plan

The plan combines two independent statistical tests on the same patients; **a CpG is “silencing” only if both tests agree with concordant direction**. This operationalizes Knudson’s tumor-suppressor second hit in the methylation–expression coupling regime: differential methylation alone cannot distinguish promoter silencing from gene-body activation, and methylation–expression coupling alone cannot distinguish a tumor-acquired event from a constitutive one.

(1) Differential methylation (DM) test — paired moderated t . For each CpG, within-pair M-value differences $D_i = M_i^{\text{tumor}} - M_i^{\text{normal}}$ ($i = 1, \dots, 160$) are tested against $H_0: \mathbb{E}[D] = 0$ using `limma::lmFit` with an intercept-only design and `eBayes` shrinkage (Smyth 2004; Ritchie et al. 2015). For balanced paired data this is equivalent to the moderated paired t -test, and patient-level confounders cancel in D_i .

(2) Methylation–expression association (MEA) test — partial-correlation regression. For each promoter (CpG, gene) pair (annotated as TSS1500, TSS200, 5'UTR, or 1stExon), we fit

$$\log_2(\text{TPM}_{\text{gene}} + 1) = \alpha + \beta \beta_{\text{CpG}} + \gamma^\top Z + \varepsilon, \quad Z = (\text{sample_type}, \text{age}, \text{sex}, \text{tissue source site}),$$

across the 342 matched samples. Adjusting for `sample_type` is essential: without it, cross-sample correlation is dominated by the bulk tumor/normal mean shift (e.g., for VHL the cross-cohort $\rho = -0.03$ vs within-status partial $p = 2 \times 10^{-6}$). The test is computed as a partial correlation after QR residualization, vectorizing to all 137,451 promoter pairs.

(3) Silencing definition. A (CpG, gene) pair is **silencing** if (i) DM FDR < 0.05 , (ii) MEA FDR < 0.05 , and (iii) $\widehat{\log\text{FC}}_{\text{DM}} > 0$ and $\hat{\beta}_{\text{MEA}} < 0$. The reverse direction (hypomethylation $\cap$ positive coupling) is recorded as **activation** and reported descriptively.

(4) Pattern partition by responder fraction. Mean-based DM dilutes signals present in only a minority of tumors — VHL, the dominant ccRCC driver, is methylation-silenced in only ~15% of tumors and is therefore invisible to cohort-mean tests. Each silencing CpG is annotated by $\hat{\pi}_{\text{resp}}$, the fraction of tumors whose β deviates from the adjacent-normal mean by > 0.2 (the standard 450K biological-significance threshold). Silencing pairs split into **subgroup-driven** ($\hat{\pi}_{\text{resp}} < 0.3$) and **cohort-wide** (≥ 0.3).

(5) Pathway enrichment and cell-composition control — central methodological claim. Bulk tumor DM is confounded by cell composition: tumor tissue carries more infiltrating leukocytes than adjacent normal kidney, so any CpG distinguishing immune from epithelial cells appears differentially methylated for compositional reasons alone. The standard fix is reference-based deconvolution (EpiDISH, CIBERSORT). Our hypothesis is that silencing’s directional concordance criterion automatically excises this signal: an immune CpG bulk-hypomethylated in tumor *by* infiltration also carries elevated bulk mRNA from those same cells — the opposite of silencing’s hypermethylation + low-mRNA signature, so it fails the concordance test. The falsifiable prediction is that immune pathways dominant in raw-DM enrichment disappear under the silencing restriction. We compare GSEA on raw-DM rankings (`fgsea`) against ORA on the silencing gene list (hypergeometric on Hallmark + KEGG Legacy).

(6) Pre-registered validation. (a) An 18-gene literature TSG list (VHL, RASSF1, SFRP1/2/4/5, DKK3, GATA5, BNC1, COL14A1, PCDH17, SLIT2, CDKN2A, APAF1, TIMP3, NEFH, UCHL1, PITX2) is the concurrent-validity sanity check. (b) **VHL positive control:** at least one VHL-promoter CpG must flag as silencing. (c) **Calibration:** the DM test’s λ_{GC} is paired with a sign-flipping permutation null on 30,000 probes $\times$ 20 replicates. (d) **External replication:** the identical pipeline is run on TCGA-KIRP (papillary RCC, 45 paired patients) and silencing-gene overlap with KIRC is tested for hypergeometric enrichment. (e) **Threshold sensitivity:** silencing counts and ORA top-hits are reported across a 3×3 (DM FDR, MEA FDR) $\in \{0.01, 0.05, 0.10\}$ grid.

(7) Multiple-testing correction. BH FDR at $q < 0.05$ is applied separately to each test; the silencing AND of two independently FDR-controlled sets is conservative relative to either single-test FDR. BH FDR is also used across gene-set families. Calibration is anchored by the permutation null in (6c) rather than by Bonferroni, which is too conservative given the pervasive real signal in this cohort.

5 Figures, tables, and feasibility

Planned outputs. Six figures: (1) PCA on top-variance CpGs ($PC1 \approx \text{tumor/normal}$, $PC2 \approx \text{sex}$); (2) DM diagnostic panel (p-value histogram with permutation-null overlay, paired- $\Delta\beta$ histogram, $\Delta\beta$ -vs-p volcano with anchor TSGs labelled); (3) MEA volcano + responder-fraction histogram; (4) six-locus methylation-expression scatter; (5) VHL/cg13672843 subgroup-driven case study; (6) raw-DM GSEA vs silencing-restricted ORA. Four tables: preprocessing attrition, immune-pathway dropout, FDR sensitivity grid, 18-gene TSG recovery.

Feasibility (pipeline implemented). A first end-to-end run of the pipeline produces the expected qualitative pattern: raw DM enrichment is dominated by immune pathways (Inflammatory Response, Complement, Interferon γ/α , $TNF\alpha/NF-\kappa B$), all of which fall out of significance under the silencing restriction, leaving recognizable cancer biology (EMT, $TGF-\beta$, KRAS, Pathways in Cancer). The VHL positive control flags as silencing, and the 18-gene literature TSG panel is enriched over baseline. External replication on TCGA-KIRP recovers a substantial silencing-gene overlap with KIRC. Quantitative results, threshold-sensitivity, and calibration are deferred to the final report.

Expected conclusion. Silencing — defined as the directional intersection of paired-design DM and within-status methylation-expression association — should be sufficient to identify transcriptionally silencing CpGs without an external cell-composition deconvolution panel. If confirmed, the immune-pathway confounding dominating raw DM collapses under the silencing restriction while cancer biology survives, and a complementary descriptive observation — that most methylation silencing is patient-subgroup-specific rather than cohort-wide — argues for routinely reporting per-tumor responder fractions in cancer methylation studies.

References

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